Feature Extraction Using PCA

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Stellar Classification

Objective

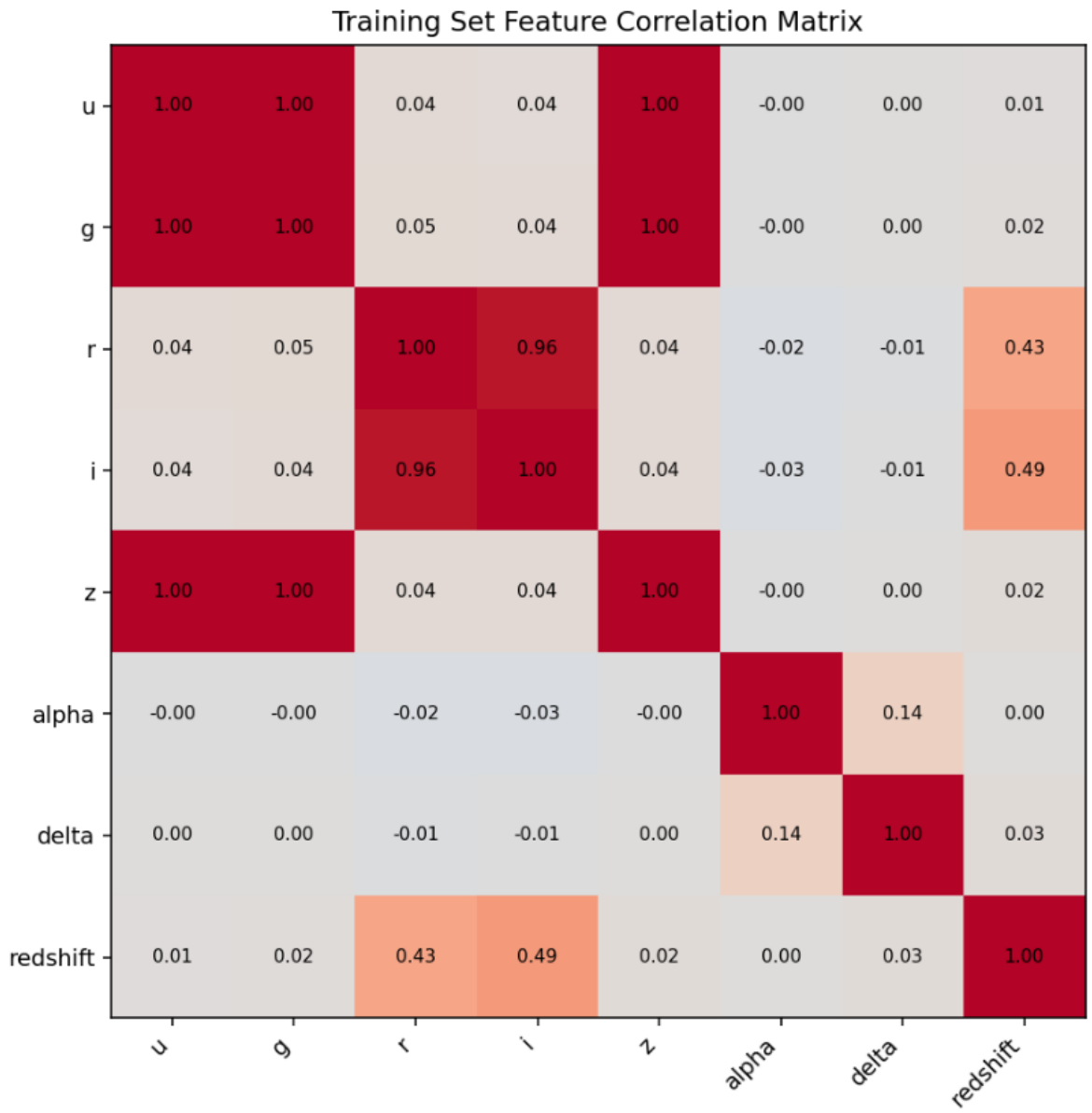
- Reduce Dimensionality of the dataset.
- Identify relationships between features.
- Improve efficiency for classification models.
- Preserve important variance in the data.

Dataset & Features

- Dataset: Sloan Digital Sky Survey (SDSS)
- Classes:
  - GALAXY, STAR, QSO
- Original Features (8):
  - U, G, R, I, Z
  - Alpha, Delta
  - Redshift

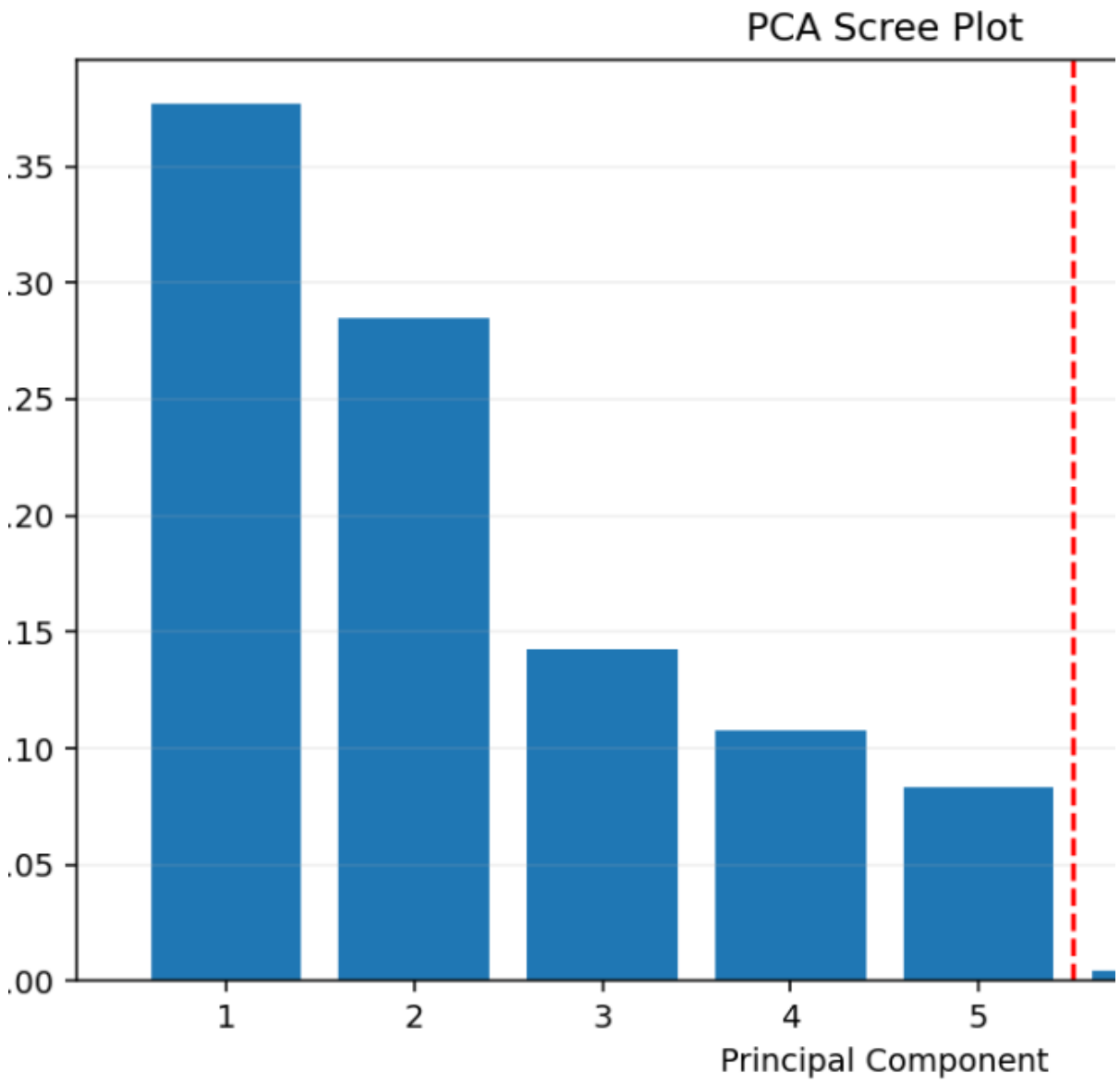
Methodology

- Computed correlation matrix on training data
- Applied PCA on training split only
- Selected components to retain ≥95% variance
- Transformed train, validation, and test datasets



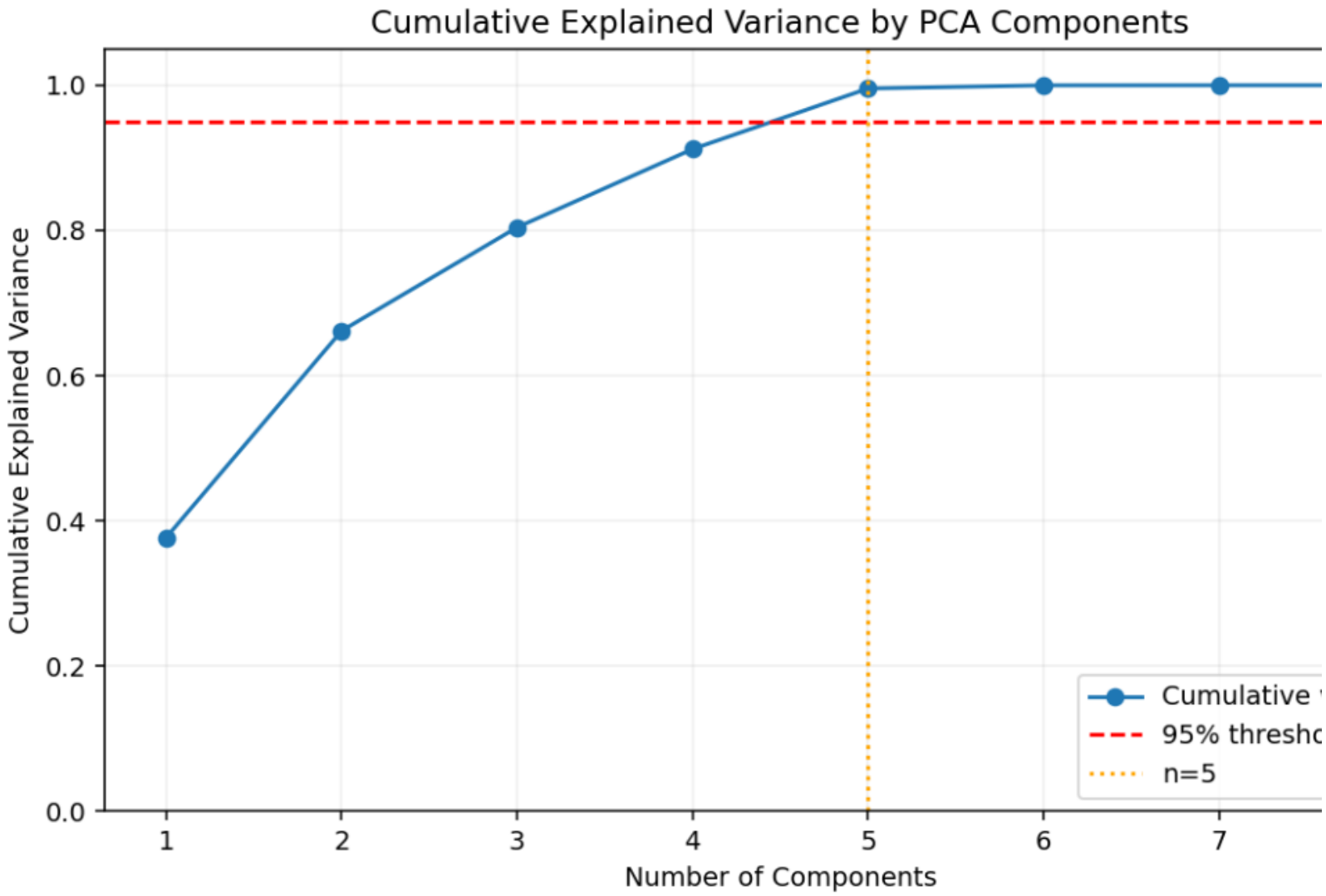
Correlation Heatmap

- Strong correlation between **u, g, z (~1.0)**
- Strong correlation between **r and I (~0.96)**
- Weak correlation with **alpha, delta**
- Indicates redundancy in original features



Scree Plot

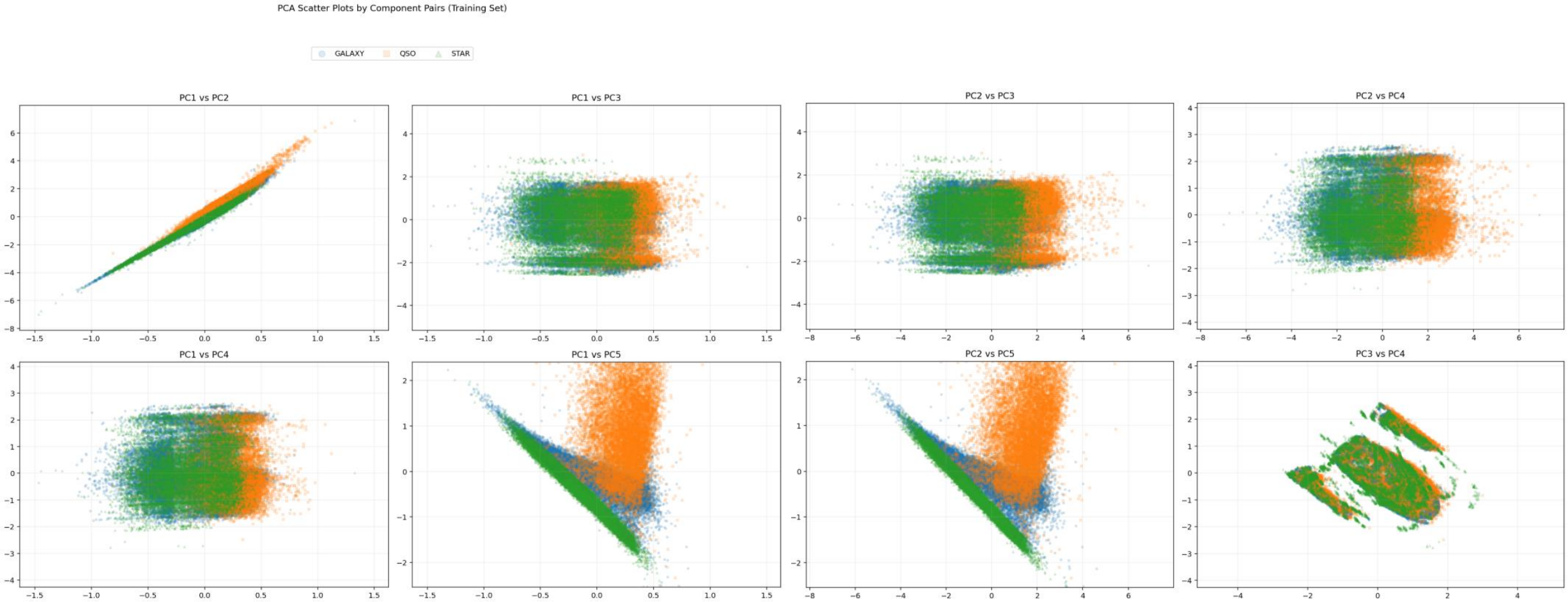
- PC1 and PC2 explain the most variance
- Sharp drop after first few components
- Minimal contribution from later components



Cumulative Variance

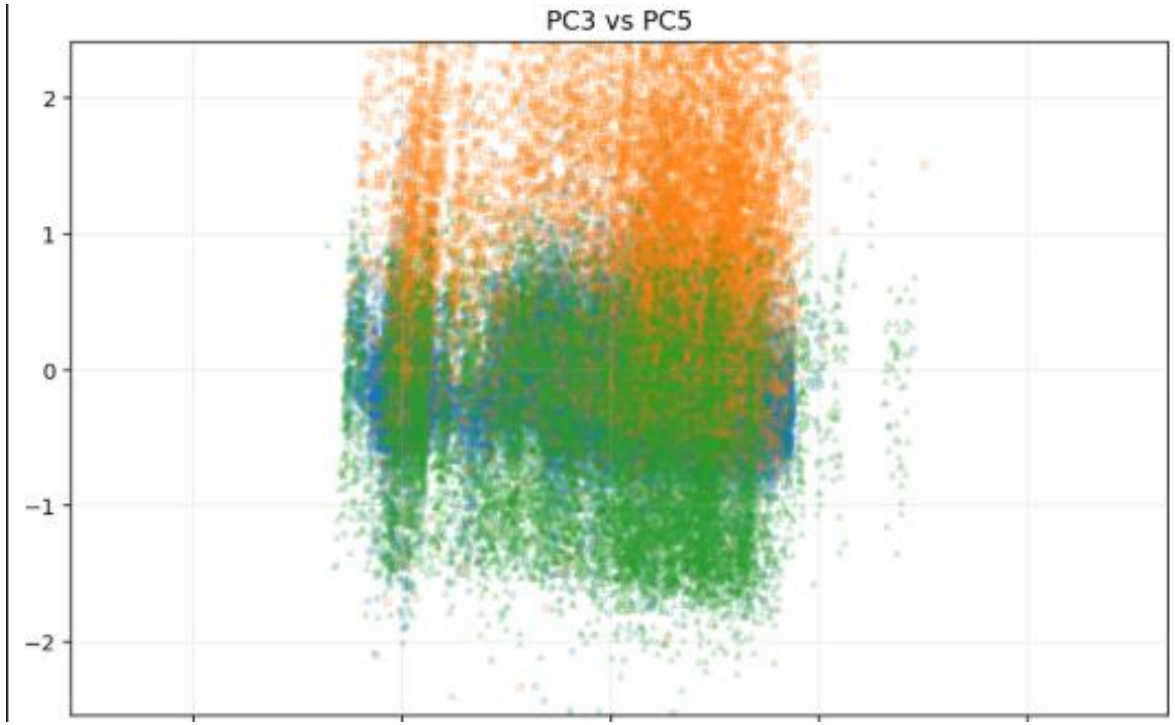
- 5 components capture ≥95% variance
- Remaining components add very little information
- Selected optimal dimensionality = 5

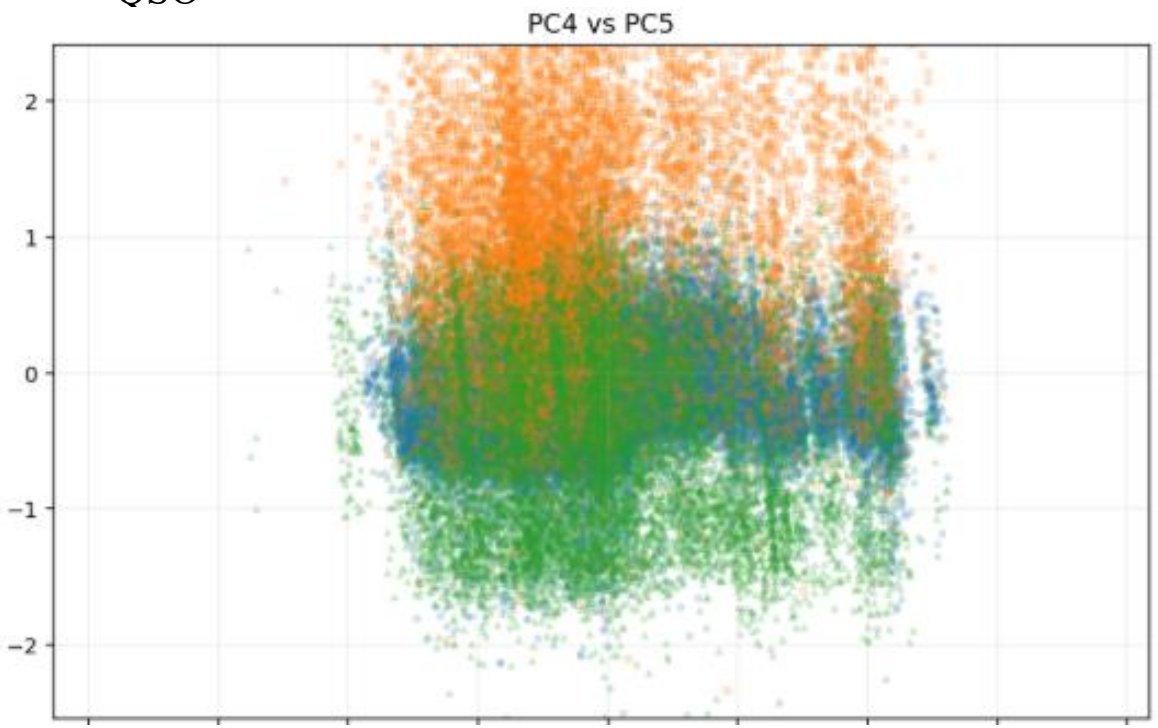
PCA Scatter Plots



PCA Scatter Plots: Continued

- Data projected into lower-dimensional space
- Improved structure compared to raw features
- Some separation between:
  - GALAXY
  - STAR
  - QSO





1	split	PC1	PC2	PC3	PC4	PC5	class
2	train	0.40569	1.841957	-0.00222	-0.42754	-0.67769	GALAXY
3	train	0.15743	0.454934	0.14999	-0.65101	-0.19148	GALAXY
4	train	0.258896	1.334261	0.299814	2.100002	-0.0923	GALAXY
5	train	-0.50213	-2.46684	1.117944	1.273476	0.521743	GALAXY
6	train	0.29147	1.448632	1.126313	1.336542	0.506451	QSO
7	train	0.148642	0.495471	0.992249	-0.63514	-0.25397	GALAXY
8	train	-0.08784	-0.56965	-0.91907	-1.1447	-0.55959	STAR
9	train	0.162394	0.459183	0.857667	1.355558	-0.44611	GALAXY
10	train	0.193801	0.686064	0.5256	0.466198	-0.3749	GALAXY
11	train	-0.04106	-0.4224	1.474773	0.966153	-0.74823	STAR
12	train	-0.33863	-1.73951	0.60378	1.598024	0.258315	GALAXY
13	train	0.3869	1.901792	0.32198	2.17627	-0.5377	GALAXY
14	train	0.124594	0.330828	0.506805	0.485345	-0.3483	GALAXY
15	train	0.180664	0.639788	0.439135	2.121753	-1.16183	STAR
16	train	0.209992	1.503986	-1.89699	-0.60339	1.615201	QSO
17	train	-0.31681	-1.64892	1.462512	-1.65193	-0.0553	GALAXY
18	train	0.139858	0.352985	0.501291	-0.98855	-1.16889	STAR
19	train	0.070663	0.059336	0.594196	-1.05364	0.004179	GALAXY
20	train	0.036942	-0.17426	-0.92883	0.412532	-0.00357	GALAXY

Extracted Features (Excel Output)

- Enables efficient input for machine learning models
- Data is now lower-dimensional but retains key structure
- Generated dataset with:
  - PC1, PC2, PC3, PC4, PC5
  - Class labels preserved
  - Train/Validation/Test splits maintained

Results

- Reduced dimensionality: **8 into 5 features using PCA**
- Retained ≥ **95%** of total variance
- Identified strong feature correlations (u, g, z and r, I)
- Successfully transformed **train, validation, and test splits**
- Generated reusable outputs:
  - PCA features (Excel)
  - PCA parameters (JSON)
  - Visualization plots for analysis

Conclusion

- PCA revealed structure and relationship in the dataset
- Reduced complexity without losing meaningful information
- Improved data representation for modeling tasks
- Demonstrates the effectiveness of dimensionality reduction in real datasets